American Computer Science League

2023-2024 ● Contest 1: Short Problem Solutions ● Elementary Division

1. Computer Number Systems

1011010_2 is converted to base 10 by taking each digit from right to left and multiplying it by increasing powers of 2. Therefore, 0*1 + 1*2 + 0*4 + 1*8 + 1*16 + 0*32 + 1*64 = 2 + 8 + 16 + 64 = 90.

D. 90

2. Computer Number Systems

To convert from base 16 or base 8 to binary, represent each hex digit with 4 binary digits and each octal digit with 3 binary digits. Therefore,

BED_{16} = 101111101101 has 3 0s;
CAF_{16} = 110010101111 has 4 0s;
7356_8 = 111011101110 has 3 0s;
2741_8 = 10111100001 has 5 0s.

Therefore, BED and 7356 have the fewest number of 0s.

E. more than 1 of them

3. Computer Number Systems

To convert 980_{10} to base 8, divide by 8 repeatedly and get the remainder each time. Read the remainders from last to first to get the number. Therefore, 980 / 8 = 122 r 4, 122 / 8 = 15 r 2, 15 / 8 = 1 r 7, 7 / 8 = 0 r 7. The remainders create the number 1724 in base 8. Notice remainders must be between 0 and 7 inclusive when dividing by 8.

B. 1724

4. Computer Number Systems

The easiest way to do this is to add the two hexadecimal numbers in base 16. 4D5C + 3FA7 = C+7=16+3, 1+5+A=16+0. 1+D+F=16+13=D, 4+3+1=8 so the answer is 8D03. Carrying a 1 is needed whenever the sum is greater than or equal to 16. Always work from right to left as in regular column addition.

D. 8D03

5. Computer Number Systems

The hex code "#241571" for the color Berry is converted to decimal for each 2-digit pair:

24 = 2*16 + 4 = 36,
15 = 1*16 + 5 = 21,
71 = 7*16 + 1 =113.

Therefore, the 3-digit decimal equivalent is (36,21,113).

C. (36,21,113)